Semiconductor Manufacturing International Corporation

Doc. No.: TD-LO40-LC-2002	Doc. Title: 40nm Low Leakage 1.1V/2.5V Calibre LVS	Doc.Rev: 1	Tech Dev. Rev: 1.15_0

Document Level: (For Engineering & Quality Document/工程暨品质文件专用)

☐ Level 1 - Manual ☑ Level 2 – Procedure/SPEC/Report ☐ Level 3 - Operation Instruction

Security Level:

☑ Security 1 - SMIC Confidential ☐ Security 2 - SMIC Restricted ☐ Security 3 - SMIC Internal

Document Change History

Doc. Rev.	Tech Dev. Rev.	Effective Date	Author	Change Description
0	1.14_0	2015-09-08	George Wang	Initial. Please be kindly noted that the contents of this document may contain the confidential information of a third party such as SMIC's technology licensor, IP vendor, or customer, etc. Pursuant to the agreement between such third party and SMIC, SMIC's employee with a need to access such third party's confidential information shall (i) use such information solely for the benefit of SMIC or the third party, and for no other purpose, (ii) keep such information confidential for a period of fifteen years or longer from its disclosure of such information to SMIC, and (iii) agree that the third party is a third party beneficiary for the confidentiality obligation stated herein.
1	1.15_0	2017-07-05	Carly Zhang	1. Modified below items in mainfile: 1) Update the document tile to “40nm Low Leakage 1.1V/2.5V Calibre LVS”in item1 according to new naming rule. 2) Update the reference document TD-LO40-SP-2001 title in item5. 3) Update the responsibility department to “Technology R&D Center /ETD/Tech-Bas Des-Enb/Compact Model 2/LVS “in item6. 4) Update the attachment name in item8.1: a. Calibre LVS runset file package to “SMIC_CalLVS_40LL_1125_V1.15_0.tar.gz” b. After decompress to “SMIC_CalLVS_40LL_1125_V1.15_0.lvs 5) Update the attachment name in item8.2 to: “SMIC_CalLVS_40LL_1125_V1.15_0.lvs.pdf” 2. Modified below items in attachment(SMIC_CalLVS_40LL_1125_V1.15_0.tar.gz): 1) Remove the switch "IO 1825/1833"; 2) Delete the 3.3v device since no model for these devices; 3) Add MARKS into ptd, add "LVS SPICE REPLICATES DEVICE:YES","LVS SPICE STRICT WL" 4) Change XRC/CCI QUERY to QUERY XRC/CCI 5) update rwd11ll area extraction from count DSTR area to rwell area 6) Standard the CONNECT part to format "connect x y by z" 7) Update dw of rnposab_2t_ckt, rnposab_3t_ckt : 1.27E-8 -> 1.265E-8 8) Add parameter "R" for resistor for PERC when BACK_ANNOTATION = 1 9) Add GT_all and replace GT by iGT for none layer definition

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				10) Add switch "PSFLAG_MOM" and set default to true. 11) Delete DONUT, STAMP from LVS to avoide hierarchy distroyed. 12) Delete unused nwellring, pw_nw ALPA_NORES, AA_SN & AA_SP. 13) Delete ntap and replace by ntd in ERC:ntd_to_ground. 14) Update the none layer definition by format: one device by on device. 15) Modify switch "XRC_RUN" to "RC_RUN". 16) Add terminal description for diode. 17) Change MOS_MODE & NTMOS_MODE to MOS_MISMODE, MOS_GLOBALMOD & MOS_LPEMOD. 18) Replace AA by iAA for none layer definition. 19) Replace MOM_POLY by ppoly for 3rd terminal of mom. 20) Delete the unused layer and connection: efpoly. 21) Delete the RF logic operation to decrease runtime. 22) Remove metal*b for all metal option and update metal4 definition for TOPMETAL =5 23) remove MOM capacitor parameters : mtl1,mtl2 24) Not ldmos terminal layer from ntd,ptd,nsd,psd to avoid RC double count. 25) Add 3.3v & 5v mom: mom33_2t_ckt/mom33_3t_ckt/mom33_4t_ckt/mom33_5t_ckt/mom50_2t_ckt mom50_3t_ckt/mom50_4t_ckt/mom50_5t_ckt 26) Optimize the SA/SB formula to tcl proc. 27) Add PDBA Flow for good performance of hierarchy keep. 28) Update mom2t/3t terminal layer to distinguish with mom array's. 29) Remove "NOT dnw" from "ndio11llulvt's seed layer". 3. Modified Below items in attached LVS release note: 1) Update document titles in item1.1. 2) Update LVS options in item1.2. 3) Update LVS mark layer in item1.3. 4) Update the device supported in item1.4. 5) Update Calibre version to 2014.3.16.15 in item1.6. 6) Update the revision history in item1.7.

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1. Title:

40nm Low Leakage 1.1V/2.5V Calibre LVS

2. Purpose:

This document is intended to be a release note for the users who adopt SMIC 40 nm Logic LowLeakage technology and apply Calibre LVS for comparing layout with Schematic for their design reference.

3. Scope:

This document presents the latest information about Calibre LVS runset for 40nm LL process.

4. Nomenclature:

LVS runset **L**ayout **V**ersus **S**chematic

5. Reference:

TD-LO40-DR-2001 40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V low leakage and generic Design Rule
TD-LO40-SP-2001 40nm Low Leakage 1.1V/2.5V SPICE Model

6. Responsibility:

Technology R&D Center /ETD/Tech-Bas Des-Enb/Compact Model 2/LVS.

7. Subject Content:

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Test Case Type	Results	Known Problems
Test Pattern	pass	NA
SMIC PDK	pass	NA

8. Attachment:

1) Calibre LVS runset file :

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SMIC_CalLVS_40LL_1125_V1.15_0.tar.gz

After decompress :
SMIC_CalLVS_40LL_1125_V1.15_0.lvs
empty_subckt.sp

2) SMIC_CalLVS_40LL_1125_V1.15_0.lvs.pdf

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				27) Add PDBA Flow for good performance of hierarchy keep. 28) Update mom2t/3t terminal layer to distinguish with mom array's. 29) Remove "NOT dnw" from "ndio11llulvt's seed layer". 3. Modified Below items in attached LVS release note: 1) Update document titles in item1.1. 2) Update LVS options in item1.2. 3) Update LVS mark layer in item1.3. 4) Update the device supported in item1.4. 5) Update Calibre version to 2014.3.16.15 in item1.6. 6) Update the revision history in item1.7.

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